

Stepper Motor Controller Command list

Target: EVO stage Production Tooling 28030-5100-911_Test_box_Axis

General

- All commands end with CR character.
- All commands should be entered in Capitals.
- Parameters separated with a space character.
- Communication USB is 115200 Baud, 8 bits, no parity.
- Motor are named X (J19), Y (J20) and Z (J21).

Command Description

SM command

Function: Set Motor mechanical parameters

Format: SM [MOTOR] [USTEPSIZE] [MOTOR_STEPS] [DIRECTION]

Returns: SM [MOTOR] [USTEPSIZE] [MOTOR_STEPS] [DIRECTION]

MOTOR = X, Y or Z

USTEPSIZE = micro step size e.g. 1, 2, 4, 16

MOTOR_STEPS = Number of motor steps per revolution e.g. 200 or 400

DIRECTION = Normal or reversed motor direction, 0 = normal, 1 = reversed.

GM command

Function: Display Motor mechanical parameters

Format: GM [MOTOR]

Returns: GM [MOTOR] [USTEPSIZE] [MOTOR_STEPS] [DIRECTION]

MOTOR = X, Y or Z

SS command

Function: Set Motor speed parameters

Format: SS [MOTOR] [MINSPEED] [MAXSPEED] [ACCEL] [DECEL] [RAMPTYPE]

Returns: SS [MOTOR] [MINSPEED] [MAXSPEED] [ACCEL] [DECEL] [RAMPTYPE]

MOTOR = X, Y or Z

MINSPEED = Start speed in 0.01 * rad/sec, range 10..10000.

MAXSPEED = Maximum motor speed in 0.01 * rad/sec, range 10..12800.

ACCEL = Acceleration in $0.01 * \text{rad/s}^2$, range 628..12800.
DECEL = Acceleration in $0.01 * \text{rad/s}^2$, range 628..12800 and $\geq$ ACCEL.
RAMPTYPE = Trapezoidal or S-Curve ramp, 0 = Trapezoidal, 1 = S-Curve.

GS command

Function: Display Motor speed parameters

Format: GS [MOTOR]

Returns: GS [MOTOR] [MINSPEED] [MAXSPEED] [ACCEL] [DECEL] [RAMPTYPE]

MOTOR = X, Y or Z

SC command

Function: Set Motor current parameters

Format: SC [MOTOR] [RUN_CURRENT] [HOLD_CURRENT]

Returns: SC [MOTOR] [RUN_CURRENT] [HOLD_CURRENT]

MOTOR = X, Y or Z

RUN_CURRENT = Motor current in mA (I_{peak}) during run, range 100..1800.

HOLD_CURRENT = Motor current in mA during hold, range 100..1800 and $\leq$ RUN_CURRENT

GC command

Function: Display Motor current parameters

Format: GC [MOTOR]

Returns: GS [MOTOR] [RUN_CURRENT] [HOLD_CURRENT]

MOTOR = X, Y or Z

FP command

Function: Flash all current Motor parameters in EEPROM. Three different sets of Speed and Peak current can be stored by given indexes.

Format: FP [MOTOR] [SPD_INDEX] {CUR_INDEX}

Returns: FP [MOTOR] [SPD_INDEX] {CUR_INDEX}

MOTOR = X, Y or Z

SPD_INDEX = Speed index to store speed parameters, range 1..3

CUR_INDEX = Current index to store motor peak current parameter, range 1..3

GP command

Function: Read all Motor parameters from EEPROM, also done at startup. Three different sets of Speed and Peak current can be readed from EEPROM by given indexes.

Format: GP [MOTOR] [SPD_INDEX] {CUR_INDEX}

Returns: GP [MOTOR] [SPD_INDEX] {CUR_INDEX}

MOTOR = X, Y or Z

SPD_INDEX = Speed index to read speed parameters, range 1..3

CUR_INDEX = Current index to read motor peak current parameter, range 1..3

PP command

Function: Print current Motor parameters.

Format: PP [MOTOR]

Returns:

Min Speed : 628 *0.01 Rad/s

Max Speed : 3140 *0.01 Rad/s

Accelleration: 3140 *0.01 Rad/s^2

Decelleration: 3140 *0.01 Rad/s^2

Ramp type : S-Curve

Max Current : 1800 mA

Hold Current : 500 mA

Microsteps : 16 Steps/Fullstep

Motor steps : 200 Fullsteps/Rev

Motor direct.: normal

MOTOR = X, Y or Z

MH command

Function: Motor Home command

Format: MH [MOTOR] [MAX_STEPS] [SENSOR_MASK] {OVERTRAVEL}

Returns: MH [MOTOR] [MAX_STEPS] [SENSOR_MASK] {OVERTRAVEL}

MOTOR = X, Y or Z

MAX_STEPS = Maximum steps and direction to find home sensor, range -30000000 ..+30000000.

If sensor was already active, motor moves until sensor deactivates.

SENSOR_MASK = Byte mask for input sensor, motor stops with OVERTRAVEL steps, range 0..255.

OVERTRAVEL = Steps to move after sensor detection,

MA command

Function: Move Motor to Absolute Position

Format: MA [MOTOR] [POSITION]

Returns: MA [MOTOR] [POSITION]

MOTOR = X, Y or Z
POSITION = Position to run to, range -30.000000 ..+30.000000.

MR command

Function: Move Motor to Relative Position

Format: MR [MOTOR] [+/-POSITION]

Returns: MR [MOTOR] [POSITION]

MOTOR = X, Y or Z
POSITION = Relative Position to run to, range -30.000000 ..+30.000000.

DP command

Function: Display Motor Position

Format: DP [MOTOR]

Returns: DP [MOTOR] [POSITION]

MOTOR = X, Y or Z
POSITION = Current motor Position.

SP command

Function: Set Motor Position

Format: SP [MOTOR] [POSITION]

Returns: SP [MOTOR] [POSITION]

MOTOR = X, Y or Z
POSITION = New Motor Position, range -30.000000 ..+30.000000.

DS command

Function: Display Motor Status

Format: DS [MOTOR]

Returns: DS [MOTOR] [STATUS]

MOTOR = X, Y or Z
STATUS = Motor status, 1 = Ready, 2 = Running.

MO command

Function: Motor off command, which switch the motor off (HOLD_CURRENT = 0).

Format: MO [MOTOR]

Returns: MO [MOTOR]

MOTOR = X, Y or Z

ME command

Function: Motor Enable command, which sets the motor to HOLD_CURRENT.

Format: ME [MOTOR]

Returns: ME [MOTOR]

MOTOR = X, Y or Z

SO command

Function: Set output bits.

Format: SO [BYTE]

Returns: SO [BYTE]

BYTE = Output bits to set, range 0..255

Meaning of Bits:

0 – Power output J1

1 – Power output J2

2 – Led 1

3 – Led 2

4 – Led 3

5 – Status output 1

6 – Status output 2

7 – Status output 3

CO command

Function: Clear output bits.

Format: CO [BYTE]

Returns: CO [BYTE]

BYTE = Output bits to clear, range 0..255

Meaning of Bits:

0 – Power output J1

1 – Power output J2

2 – Led 1

3 – Led 2

- 4 – Led 3
- 5 – Status output 1
- 6 – Status output 2
- 7 – Status output 3

GO command

Function: Display output byte.

Format: GO

Returns: GO [BYTE]

BYTE = Output byte, range 0..255

Meaning of Bits:

- 0 – Power output J1
- 1 – Power output J2
- 2 – Led 1
- 3 – Led 2
- 4 – Led 3
- 5 – Status output 1
- 6 – Status output 2
- 7 – Status output 3

GI command

Function: Display value of 8 Input sensors.

Format: GI

Returns: GI [BYTE]

BYTE = Output byte, range 0..255

Meaning of Bits:

- 0 – Sensor 1
- .
- .
- 7 – Sensor 7

DM command

Function: Display test module type. To be set with dipswitch on Main board.

Format: DM

Returns: DM [BYTE]

BYTE = Output byte, range 0..2

0 = XY test box (SW 1 = ON, SW 2 & SW 3 = OFF)
1 = Z test box (SW 2 = ON, SW 1 & SW 3 = OFF)
2 = RT test Box (SW 3 = ON, SW 1 & SW 2 = OFF)

VE command

Function: Display firmware version.

Format: VEI

Returns: '* Stepper Motor Controller Vx.xx *'

RESET command

Function: Resets the Controller.

Format: RESET

Returns: '* Stepper Motor Controller Vx.xx *'

? command

Function: Help command to display available commands.